D Greek Letters and Mathematical Symbols Used in the Text

Greek letters and common interpretation

alpha	α	population linear projection coefficient (page 74)
beta	$\boldsymbol{\beta}$	population regression coefficient (page 200)
gamma	$\boldsymbol{\Gamma}$	autocovariance matrix for vector process (page 261)
	γ	autocovariance for scalar process (page 45)
delta	Δ	change in value of variable (page 436)
	δ	small number; coefficient on time trend (page 435)
epsilon	ε	a white noise variable (page 47)
zeta	ζ	constant term in *ARCH* specification (page 658)
eta	η	$AR(\infty)$ coefficient (page 79)
theta	$\boldsymbol{\Theta}$	matrix of moving average coefficients (page 262)
	$\boldsymbol{\theta}$	vector of population parameters (page 747)
	θ	scalar $MA(q)$ coefficient (page 50)
kappa	κ	kernel (page 165)
lambda	$\boldsymbol{\Lambda}$	matrix of eigenvalues (page 730)
	λ	individual eigenvalue (page 729) Lagrange multiplier (page 135)
mu	μ	population mean (page 739)
nu	ν	degrees of freedom (page 409)
xi	$\boldsymbol{\Xi}$	matrix of derivatives (page 339)
	$\boldsymbol{\xi}$	state vector (pages 7 and 372)
pi	Π	product (page 747)
	π	the number 3.14159....
rho	ρ	autocorrelation (page 49) autoregressive coefficient (page 517)
sigma	$\sum$	summation
	$\boldsymbol{\Sigma}$	long-run variance-covariance matrix (page 614)
	σ	population standard deviation (page 745)
tau	τ	time index
upsilon	$\boldsymbol{\Upsilon}$	scaling matrix to calculate asymptotic distributions (page 457)
phi	$\boldsymbol{\Phi}$	matrix of autoregressive coefficients (page 257)
	ϕ	scalar autoregressive coefficient (page 53)
chi	χ	a variable with a chi-square distribution (page 746)
psi	$\boldsymbol{\Psi}$	matrix of moving average coefficients for vector $MA(\infty)$ process (page 262)
	ψ	moving average coefficient for scalar $MA(\infty)$ process (page 52)
omega	$\boldsymbol{\Omega}$	variance-covariance matrix (page 748)
	ω	frequency (page 153)

Common uses of other letters

a	number of elements of unknown parameter vector (page 135)
$\mathbf{b}$ or $\mathbf{b}_T$	estimated *OLS* regression coefficient based on sample of size T (page 75)
$\mathbf{c}$	vector of constant terms for vector autoregression (page 257)
c	constant term in univariate autoregression (page 53)
$\mathbf{e}_i$	the ith column of the identity matrix
e	the base for the natural logarithms (page 715)
$\mathbf{I}_n$	the $(n \times n)$ identity matrix (page 722)
i	the square root of negative one (page 708)
J	value of Lagrangean (page 135)
k	the number of explanatory variables in a regression
L	the lag operator (page 26)
$\mathscr{L}$	value of log likelihood function (page 747)
n	number of variables observed at date t in a vector system (page 257)
$O_p(T)$	order T in probability (page 460)
$\mathbf{P}_{\tau\|t}$	*MSE* matrix for inference about state vector (page 378)
p	the order of an autoregressive process (page 58)
$\mathbf{Q}$	limiting value of $(\mathbf{X}'\mathbf{X}/T)$ for $\mathbf{X}$ the $(T \times k)$ matrix of explanatory variables for an *OLS* regression (page 208); variance-covariance matrix of disturbances in state equation (page 373)
q	the order of a moving average process (page 50); number of autocovariances used in Newey-West estimate (page 281)
$\mathbf{R}$	variance-covariance matrix of disturbances in observation equation (page 373)
$\mathbb{R}^n$	the set consisting of all real n-dimensional vectors (page 737)
r	number of variables in state equation (page 372); index of date for a continuous-time process
s^2 or s_T^2	unbiased estimate of residual variance for an *OLS* regression with sample of size T (page 203)
s_t	state at date t for a Markov chain
T	the number of dates included in a sample
$\mathbf{X}$	$(T \times k)$ matrix of explanatory variables for an *OLS* regression (page 201)
$\mathcal{Y}_t$	history of observations through date t (page 143)
z	argument of autocovariance generating function (page 61)

Mathematical symbols

$\aleph$	aleph (first letter of the Hebrew alphabet), used for matrix of regression coefficients (page 636)
$\exp(x)$	the number e (the base for natural logarithms) raised to the x power (page 715)
$\log(x)$	natural logarithm of x (page 717)
$x!$	x factorial (page 713)
$[x(L)]_+$	annihilation operator (page 78)
$[x]^*$	greatest integer less than or equal to x
$\|x\|$	absolute value of a real scalar or modulus of a complex scalar x (page 709)
$\|\mathbf{X}\|$	determinant of a square matrix $\mathbf{X}$ (page 724)
$\mathbf{X}'$	transpose of the matrix $\mathbf{X}$ (page 723)
$\mathbf{0}_{nm}$	an $(n \times m)$ matrix of zeros
$\mathbf{0}'$	a $(1 \times n)$ row vector of zeros
∇	gradient vector (page 735)
$\otimes$	Kronecker product (page 732)

$\odot$	element-by-element multiplication (page 692)
$y \cong x$	y is approximately equal to x
$y \equiv x$	y is defined to be the value represented by x
$\max\{y,x\}$	the value given by the larger of y or x
$y = \sup_{r\in[0,1]} f(r)$	y is the smallest number such that $y \geq f(r)$ for all r in $[0,1]$ (page 481)
$x \in A$	x is an element of A
$A \subset B$	A is a subset of B (page 189)
$P\{\mathrm{A}\}$	probability that event A occurs (page 739)
$f_Y(y)$	probability density of the random variable Y (page 739)
$Y \sim N(\mu,\sigma^2)$	Y has a $N(\mu,\sigma^2)$ distribution (page 745)
$Y \approx N(\mu,\sigma^2$	Y has a distribution that is approximately $N(\mu,\sigma^2)$ (page 210)
$E(X)$	expectation of X (page 740)
$\mathrm{Var}(X)$	variance of X (page 740)
$\mathrm{Cov}(X,Y)$	covariance between X and Y (page 742)
$\mathrm{Corr}(X,Y)$	correlation between X and Y (page 743)
$Y\|X$	Y conditional on X (page 741)
$\hat{P}(Y\|X)$	linear projection of Y on X (pages 74–75)
$\hat{E}(Y\|X)$	linear projection of Y on X and a constant (pages 74–75)
$\hat{y}_{t+s\|t}$	linear projection of y_{t+s} on a constant and a set of variables observed at date t (page 74)
$x_T \to y$	$\lim_{T\to\infty} x_T = y$ (page 180)
$x_T \xrightarrow{p} y$	x_T converges in probability to y (pages 181, 749)
$x_T \xrightarrow{m.s.} y$	x_T converges in mean square to y (pages 182, 749)
$x_T \xrightarrow{L} y$	x_T converges in distribution to y (page 184)
$x_T(\cdot) \xrightarrow{p} x(\cdot)$	the sequence of functions whose value at r is $x_T(r)$ converges in probability to the function whose value at r is $x(r)$ (page 481)
$x_T(\cdot) \xrightarrow{L} x(\cdot)$	the sequence of functions whose value at r is $x_T(r)$ converges in probability law to the function whose value at r is $x(r)$ (page 481)